



Ottimizzazione di nuove strategie per la diagnosi della Distrofia Miotonica

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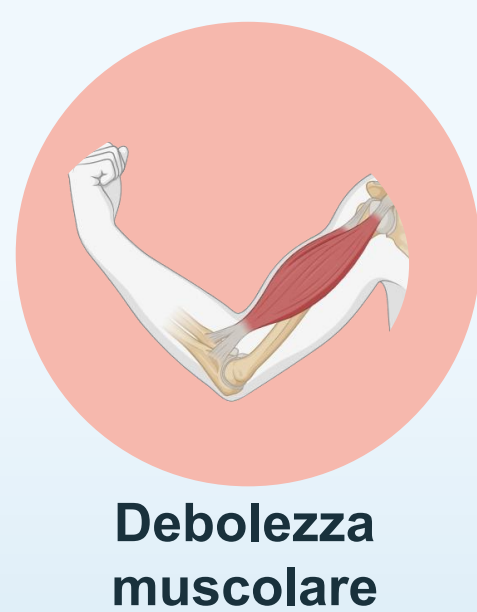
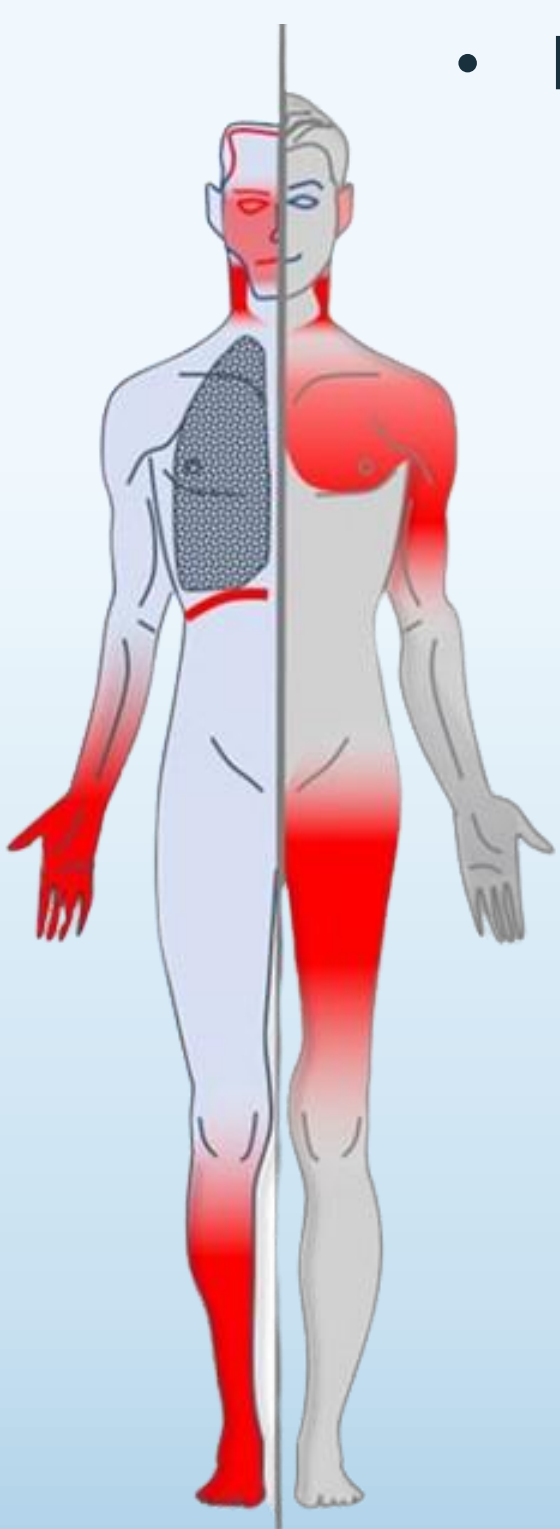


Il DNA che si ripete

1 Cos'è la Distrofia Miotonica?

La distrofia miotonica è una malattia...

- Genetica ereditaria
- Causata da ripetizioni espanse
- Rara: circa 1 persona su 8.000
- La più comune distrofia muscolare dell'adulto



Debolezza
muscolare



Miotonia



I pazienti impiegano
più di un minuto per
aprire la mano



Coinvolgimento
cardiaco



Altri sintomi

2 La Causa Genetica

Le ripetizioni espanse del DNA sono sequenze di pochi nucleotidi (lettere), che si possono **ripetere in modo anomalo**



Sano



Il senso si capisce



Malato



Il senso si perde



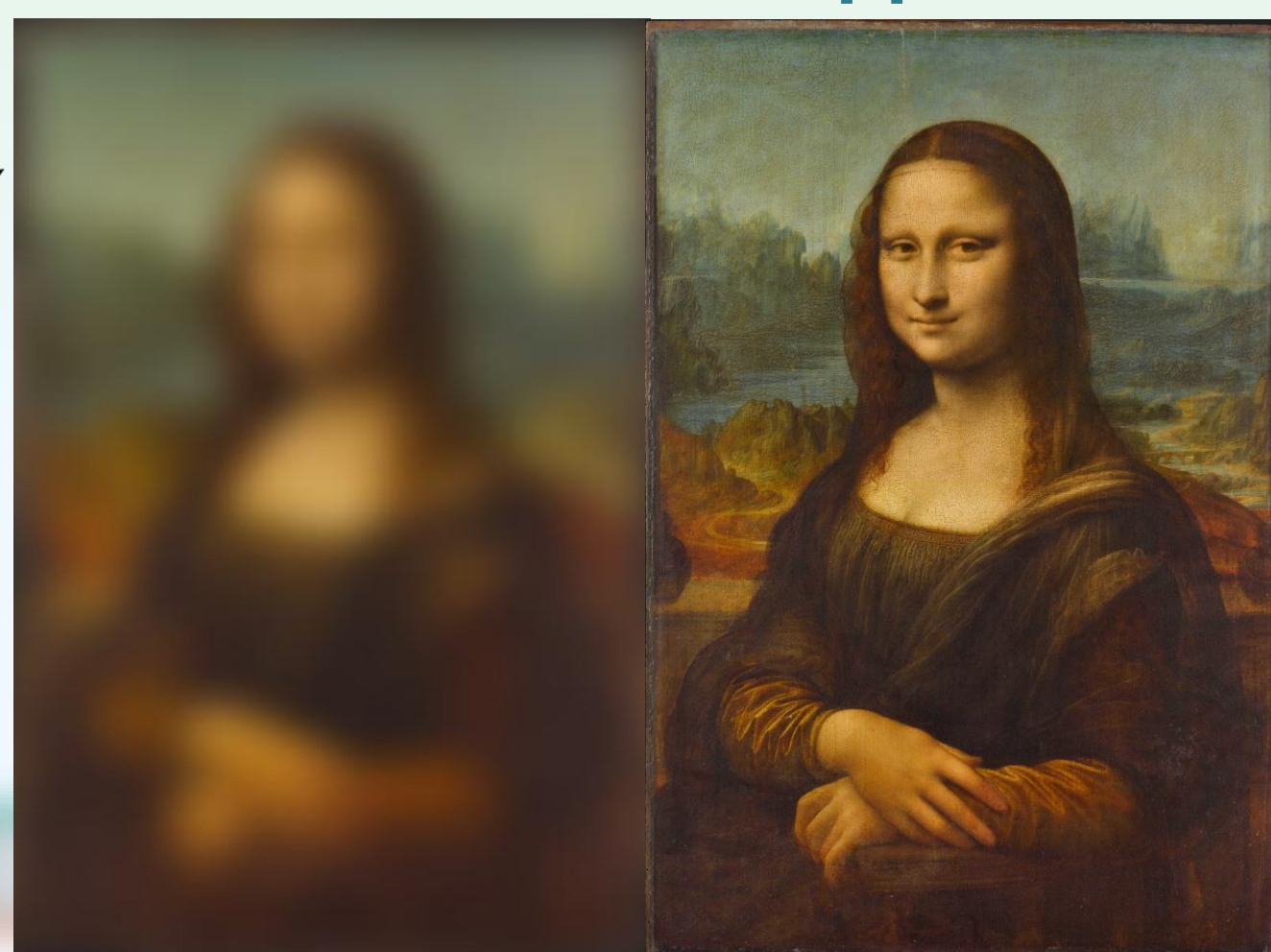
Quando le "parole" si ripetono oltre un certo limite, si sviluppa la malattia

3 Metodi diagnostici

Metodi
tradizionali

Il nostro
approccio

Bassa
risoluzione e
accuratezza



Alta
risoluzione e
accuratezza



Identificare l'espansione non basta, vogliamo vedere com'è fatta

4 Il metodo in sviluppo in ENTRY-DM

Campione



DNA



Sequenziamento



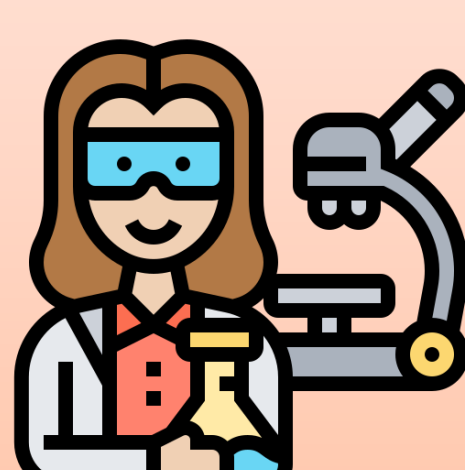
Il nostro lavoro: ottimizzare e validare ogni passaggio

5 Dal DNA del paziente alla terapia

Terapia personalizzata



Ricerca



Collegiamo il DNA ai
sintomi



Analisi della ripetizione



Take-home message

La stessa diagnosi può manifestarsi in modo molto diverso.
La risposta può essere nelle ripetizioni del DNA... e oggi sappiamo leggerle.



Funded by
the European Union

Funded by the European Union under the project
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to adapt the image
bc i literally stole it
from mario

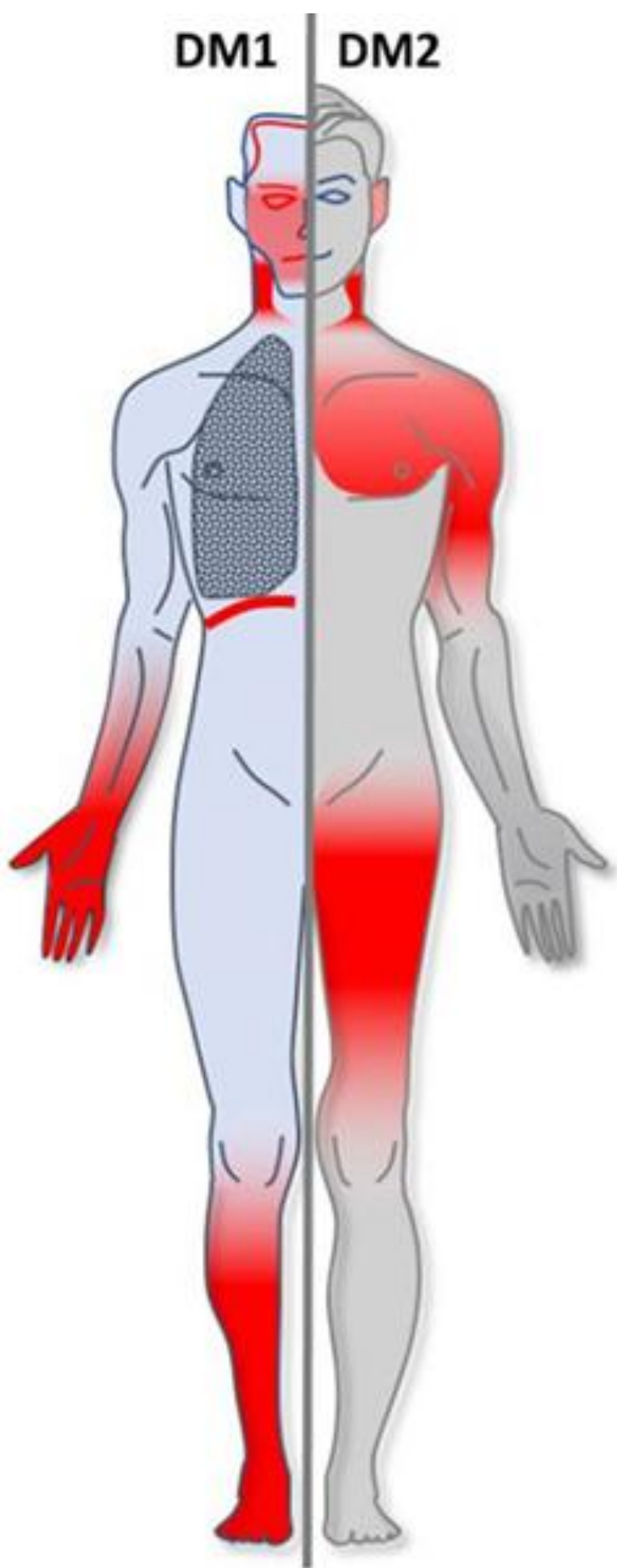


Is an european doctoral network with the purpose of training a new generation of researchers to push the frontiers of therapies development and technology transfer, creating a dynamic and collaborative training environment



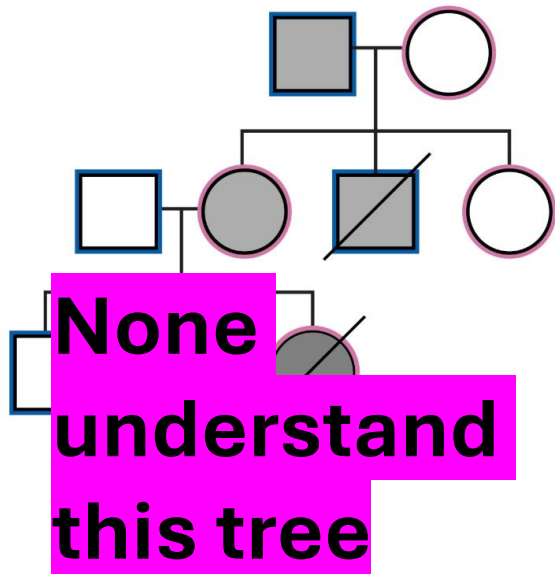
Meet out DCs!

Myotonic Dystrophy



1 What is DM?

Myotonic Dystrophy is an inherited genetic disorder, being the most common form of muscular dystrophy in adults



2 SYMPTOMS

- Muscle weakness and inability for muscles to relax quickly after contraction



One of the signs of the disease is the inability to open your hand in less than... 1 minute

3 A MULTISYSTEMIC DISEASE

Eye Problems: Clouding of the eye lens, known as cataracts, often develops early.

Heart Issues: Irregular heartbeats or rhythm defects.

Other Effects: Digestive trouble, daytime sleepiness, and hormonal imbalances

TOO MANY WORDS!!!

The causes of the disease



Ikea instructions is a nice idea, but I would show instructions correct vs incorrect in parallel to make it more efficient

Imagine a book of instructions to make all the furniture of your home...(That's your DNA!)

Imagine a specific chapter, instructions to make a specific furniture (your genes!)

But... what would happen if a **TOO MANY WORDS!!!** repetition of words?

If the phrase "tighten the screws" keeps repeating over and over, at some point it will break... **Leading to a disease**



Healthy
CTG CTG



Affected
CTG CTG



How we analyze the repeat

2 Capture the expansion



To explain here:

Our job here is to improve each step...

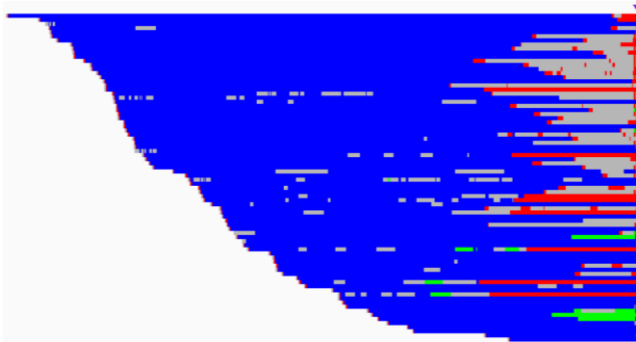
and validate it...

To bring new approaches to clinics...

3 Read it with our sequencers (a very small one!)



4 Reconstruct the expansion

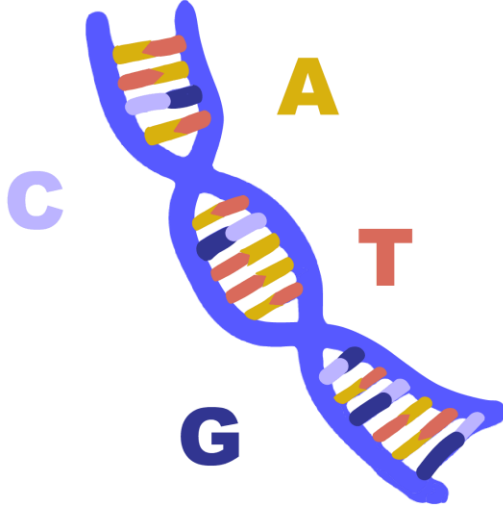


blue: CCTG red: TCTG

But what is all this info used for?

Normal approaches are good, but they don't tell you the whole story!

DNA HAS THE FULL STORY, WE READ IT



How long is the expansion?
Which Word is repeated?
All the lines has mutations?
Is there any epigenetic change?

COINNECT THE DNA TO THE SYMPTOMS



Whats the correlation with the symptoms of the patient?

CLINICS USE THE INFO



Better genetic tests
Better stratification
Personalized therapies

**Make it more effective
mentiong also therapies**